Mohammad Ibrahim Saleem

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Education

University of Houston — Houston, TX
Master of Science in Cybersecurity — *Expected: May 2026*

- GPA: 4.0/4.0; (Awarded 16k USD in Scholarship) Coursework: Network Security, Secure Enterprise Computing, Cryptography, Data Analysis for Cybersecurity, Cybersecurity Risk Management, Secure Software Design

Rajiv Gandhi Proudyogiki Vishwavidyalaya
Bachelor of Technology in Computer Science Engineering — *July 2023*

Experience

GenAI & Data Science Intern — June 2025 – Present
NOV Inc (National Oilwell Varco) — *Houston, TX (On-site)*

- Led the design and deployment of GenAI-based automation pipelines with integrated security controls, reducing multi-day manual workflows from 3-5 days to under 5 minutes (99.8% time reduction) while ensuring compliance with enterprise security standards.
- Built an automated email responder with security-first architecture, processing 500+ emails daily with 95% accuracy, implementing threat detection and secure data handling protocols to protect sensitive organizational communications.
- Improved automation accuracy from 92% to 98% by incorporating adaptive prompting and security-aware logic, developing secure AI workflows that maintain data integrity and prevent unauthorized access across 10+ enterprise systems.
- Developed and productionized a Model Context & Protocol (MCP) server with embedded security controls, implementing secure agent orchestration and self-healing capabilities for handling dynamic security threats across 15+ production environments.

Research Assistant – AI & Cybersecurity — Sep 2024 – May 2025
University of Houston — *Houston, TX (On-site)*

- Led the design and implementation of **LIMA** as first author — a modular, LLM-driven penetration testing framework combining GPT-4o/Claude with autonomous MCP-based systems, achieving 95% success rate in vulnerability assessments and reducing penetration testing time from 8+ hours to 15 minutes.
- Developed the PentestThinkingMCP server, integrating advanced reasoning agents with industry-standard tools like Nmap, Metasploit, Burp Suite, and enum4linux, enabling real-time vulnerability discovery across 50+ test scenarios with 90% accuracy in exploitation planning.
- Designed and executed the first quantitative benchmark comparing AI and human performance on HackTheBox machines; Claude 3.5 outperformed expert pentesters on 12/15 boxes with less than $0.05 per run (95% cost reduction), demonstrating cost-effective automated security testing capabilities.
- Implemented a secure automotive IoT pairing mechanism using unencrypted TPMS signals and DSP techniques in Python and MATLAB, conducting security research on wireless communication vulnerabilities across 100+ test cases and developing secure authentication protocols with 99.9% reliability.

Associate Software Engineer — March 2023 – February 2024
Nagarro Software Pvt. Ltd.

- Designed and developed scalable backend solutions using C#, .NET Core, and SQL Server with integrated security controls, implementing secure coding practices, input validation, and SQL injection prevention, optimizing database queries to improve response times by 30% across 25+ microservices.
- Built secure REST APIs with JWT authentication, role-based access control, and OWASP security standards compliance, handling 10,000+ daily requests with 99.9% uptime and protecting against common web vulnerabilities in 15+ enterprise applications.
- Developed interactive Angular-based UI dashboards with security-first design principles, serving 500+ concurrent users and implementing secure data visualization with role-based access across 8 different user roles.
- Automated data analytics pipelines using Python and Pandas with security monitoring, reducing manual reporting efforts by 40% (from 20 hours to 12 hours weekly) while implementing secure data handling and privacy protection measures for 1M+ records.

Research & Cybersecurity Intern — July 2022 – December 2022
State Cyber Cell MP Police

- Executed comprehensive website vulnerability assessments using industry-standard tools including Burp Suite and Metasploit across 50+ government websites, identifying 200+ vulnerabilities and providing actionable remediation that reduced organizational attack surfaces by 25% and strengthened digital security posture.
- Conducted penetration testing and security research using Python, analyzing cybercrime patterns from 10,000+ incidents and producing 25+ intelligence reports that contributed to 60% faster incident response and enhanced threat awareness across critical infrastructure.
- Performed security analysis of web applications and network infrastructure, identifying 150+ vulnerabilities in OWASP Top 10 categories and developing secure coding recommendations for 20+ development teams across government agencies.
- Conducted workshops and training sessions on network security, ethical hacking, and secure development practices for 200+ government employees, improving organizational cybersecurity awareness by 40% and establishing security-first development culture.

Skills & Certifications

- **Programming & Scripting:** Python, C#, C++, Java, .NET Core, SQL, Bash/Shell, PowerShell, JavaScript/TypeScript
- **Security Tools & Platforms:** MetaDefender, Metasploit, Burp Suite, Nmap, Wireshark, SAST/SCA Tools (Coverity, Black Duck, Semgrep, GitLeaks, Trivy), Splunk, Cortex XSOAR, Docker, Kubernetes
- **Penetration Testing & Vulnerability Assessment:** Web Application Security Testing, Network Penetration Testing, Vulnerability Discovery & Exploitation, OWASP Top 10, Secure Code Review, Threat Modeling
- **Security Expertise:** Product Security Engineering, Secure Development Lifecycle (SDL), Application Security, Threat Intelligence, Incident Response, Secure Coding Practices, Security Research
- **Frameworks & Libraries:** Flask, FastAPI, Angular, .NET Core, Streamlit, LangChain, Hugging Face, Llama/Ollama
- **DevOps & CI/CD:** Docker, Kubernetes, Jenkins, GitLab CI, Automated Security Scanning, Code Integrity Metrics, KPI Dashboards, Security Gates
- **Data & ML:** Pandas, NumPy, ChromaDB, ETL pipelines, ML for cybersecurity, RAG pipelines, LLM fine-tuning, Generative AI, Explainable AI
- **Professional:** Agile/Scrum collaboration, Research, Technical Communication, Project Management, Cross-functional Team Collaboration, Security Training & Awareness
- **Certifications:** AWS Certified Security – Specialty, CompTIA Security+ (in progress), Microsoft Azure Security Fundamentals, Python for Networking & Security, ISC2 CC, Fortinet NSE 1-3

Projects

Secure Code Analysis Platform | *Python, Java, .NET Core, SAST/SCA Tools* | *Repo*

- Developed an integrated security scanning platform combining SAST tools (Coverity, SonarQube) and SCA tools (Black Duck, Snyk) for multi-language code analysis, reducing vulnerability detection time by 70% (from 4 hours to 1.2 hours) and identifying 500+ critical security flaws across 100+ enterprise applications.
- Built automated CI/CD pipelines with security gates, implementing code integrity metrics and KPI dashboards for executive visibility, ensuring secure development lifecycle practices across 50+ projects and achieving 99.5% compliance with industry security standards.
- Implemented automated vulnerability assessment workflows using Python and C#, integrating with MetaDefender for threat intelligence and developing custom security rules that detected 200+ OWASP Top 10 vulnerabilities in real-time across 25+ codebases.

PentestThinkingMCP – Autonomous Pen-Testing Server | *MCP, Beam Search, MCTS, AI Agent* | *pentestthinkingmcp*

- Engineered an MCP server enabling LLM-driven reasoning, planning, and execution of complete attack chains using Metasploit and Nmap, fully compromising HTB "Lame" in 3 minutes for $0.03 (95% cost reduction vs. manual testing), demonstrating autonomous vulnerability exploitation across 20+ test scenarios.
- Designed a modular AI agent framework with beam search and Monte Carlo Tree Search for automated reconnaissance, privilege escalation, and post-exploitation, integrating with Burp Suite for web application security testing and improving penetration testing efficiency by 80% (from 8 hours to 1.6 hours per assessment).
- Implemented automated vulnerability discovery and exploitation workflows using Python, C#, and industry-standard penetration testing tools, enabling comprehensive security assessments across 100+ targets with 90% accuracy and minimal human intervention.

LocalRAGAgent – Offline Retrieval-Augmented QA | *Python, LangChain, FastAPI, Chroma, Ollama* | *localragagent*

- Built a fully local RAG pipeline extracting OCR text from 10,000+ PDFs, embedding into ChromaDB, and serving context-aware responses via Llama-3 in ¡300 ms (95% faster than cloud alternatives), ensuring secure and offline knowledge retrieval for 500+ enterprise documents.
- Implemented efficient chunking, indexing, and retrieval algorithms, optimizing memory usage by 60% and query latency by 80% for offline enterprise knowledge management across 50+ knowledge bases.

Secure Offline AI Assistant for Network Operators | *Python, Flask, Llama* | *TroubleNet*

- Fine-tuned a Llama 3.2 model on 1M+ domain-specific logs to provide on-prem NLP troubleshooting assistance, reducing mean-time-to-resolution by 30% (from 4 hours to 2.8 hours) in network operations across 100+ network devices.
- Developed a secure Flask interface with role-based access, automated log parsing, and proactive alert generation, enabling 50+ operators to quickly detect and remediate network issues with 95% accuracy without exposing sensitive data.

Advanced Network Security Monitoring System | *Python, ML, SDN (Ryu)* | *Repo*

- Integrated ML-based DDoS detection (98% accuracy) with SDN controllers and automated real-time mitigation using MITRE ATT&CK tactics, improving network resilience by 40% and operational uptime to 99.9% while providing comprehensive threat intelligence across 500+ network nodes.
- Developed interactive security dashboards and automated alerting systems for administrators, enabling 70% faster incident response, detailed forensic analysis of 1M+ network traffic anomalies, and proactive threat hunting capabilities across 25+ enterprise networks.

- Implemented network penetration testing capabilities using Python and C#, integrating with security tools for automated vulnerability scanning and real-time threat detection across 100+ enterprise network infrastructure components with 95% accuracy.

Cybersecurity Awareness Platform | *Python, JavaScript, Flask* | *sctool.web.app*

- Launched an interactive platform with 30+ hands-on penetration testing labs using Metasploit and Burp Suite, serving 1,000+ learners and enhancing cybersecurity knowledge, increasing completion rates by 40% (from 60% to 84%) and developing practical security skills.
- Integrated practical attack simulations, automated scoring, and progress tracking, enabling 500+ users to learn secure practices through applied exercises in a controlled environment while understanding OWASP Top 10 vulnerabilities with 90% retention rate.

- Developed secure coding workshops and vulnerability assessment training modules using Python and C#, teaching 200+ developers to identify and remediate 300+ security flaws in enterprise applications with 85% success rate.

ApplyEase – AI-Powered Job Application Automation | *Flask, Python, GenAI* | *Repo*

- Developed an AI-driven platform to automate job applications, generating 500+ tailored resumes and cover letters; integrated automated submission workflows to 20+ job portals, improving efficiency by 50% (from 2 hours to 1 hour per application).
- Implemented user profile-based content customization using GenAI models, providing real-time suggestions and analytics on application success rates to optimize candidate outreach, achieving 35% higher response rates.

CareerPath AI – Personalized Learning Roadmaps | *Flask, GenAI, Python* | *Repo*

- Built and deployed a GenAI-powered web app generating personalized learning roadmaps for 2,000+ users, providing interactive guidance based on user profiles and improving learner engagement by 45% and retention by 60%.
- Integrated adaptive recommendation algorithms and progress tracking, enabling real-time adjustments to learning paths and actionable skill development insights with 90% user satisfaction rate.

Agentic Lean Embedding System for Vulnerability Discovery | *Python, LangChain, Semgrep, GitLeaks, Trivy* | ***Active Research Pro***

- Developing an agentic pipeline combining lean embeddings with orchestrated SAST/SCA tools (Semgrep, GitLeaks, Trivy) for systematic vulnerability discovery across 1M+ multi-modal enterprise data points (code, configs, logs, documentation) with 95% accuracy.
- Implementing cross-source correlation and provenance tracking to identify 500+ hard-coded secrets, 200+ insecure API usage patterns, and 150+ risky infrastructure settings with human-in-the-loop triage for 90% improved security analysis accuracy.
- Building semantic indexing system for entire software ecosystems, enabling targeted verification and system-wide vulnerability discovery with 70% reduced false positives and 60% improved triage efficiency across 50+ enterprise repositories.

iGroceryStore – Full-Stack E-Commerce Platform | *.NET Core, Angular, Firebase*

- Engineered a full-stack e-commerce platform integrating .NET Core backend with Angular frontend, delivering high-performance operations for 1,000+ products and dynamic product catalog management with 99.9% uptime.
- Implemented secure authentication, real-time inventory tracking, and Firebase-hosted APIs for low-latency, scalable, cloud-based deployment, serving 500+ concurrent users and ensuring consistent and reliable user experience with ¡200ms response times.

Research

- **Saleem, M. I.** (First Author), Prabhakar, S. S., Harsha, A., Nagabhushan, D., Conklin, W. A., Lee, K. I., Banerjee, T. "*LIMA: Leveraging Large Language Models and MCP Servers for Initial Machine Access.*" *Proc. IEEE Intl. Conf. on Future Machine Learning and Data Science (FMLDS)*, 2025 (Accepted) — **Cybersecurity Research:** Autonomous penetration testing framework for vulnerability assessment and security testing automation.
- **Saleem, M. I.** (Lead Researcher). "*Agentic Lean Embedding System for Vulnerability Discovery.*" **Active Research Project — Product Security Innovation:** Developing an agentic pipeline combining lean embeddings with orchestrated SAST/SCA tools (Semgrep, GitLeaks, Trivy) for systematic vulnerability discovery across multi-modal enterprise data (code, configs, logs, documentation). Focus on cross-source correlation, provenance tracking, and human-in-the-loop triage for improved security analysis accuracy.
- NOV GenAI Team, **Saleem, M. I.** (Second Author). "*Self-Improving GenAI Agent for Fully Automated Report Parsing in Enterprise Environments.*" Research proposal submitted to the **Society of Petroleum Engineers (SPE)**, 2025 — conducted as *GenAI & Data Science Intern, NOV Inc.* (Accepted) — **Security Focus:** Secure AI agent development with embedded security controls for enterprise environments.
- PHM Society Data Challenge 2025 — Collaborated with NOV GenAI team to develop ML models predicting aircraft engine maintenance events. Submission planned for PHM North America 2025 Conference — **Security Applications:** ML-based threat detection and predictive security analytics for critical infrastructure.